

Predicting Volatility in Exchange-Traded Funds

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CERTIFICATE

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DECLARATION

I, **SHREYAN KAUL**, student of 10th semester, Five Years Integrated M.Sc. Economics (Hons.) Program of batch 2020-25, hereby attest that the dissertation I turned in, "**Predicting the Volatility in Exchange Traded Funds**" submitted under the direction and supervision of Dr. Gopinathan R. is an authentic research project that is also free of plagiarism. Additionally, I affirm that it has never before been presented, in whole or in part, to this university or institute for the awarding of a degree or diploma. Also included is a report on plagiarism statistics from this university. Additionally, I did not prepare my thesis using any artificial intelligence (AI) tool-produced content. I further stated that I will bear full responsibility for any future instances of plagiarism via AI.

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ABSTRACT

The study explores the volatility forecasting for an ETF called NIFITETF, which is the ticker symbol or short code for the actual name of the ETF namely, UTI MUTUAL FUND – UTI NIFTY IT ETF. We have collected the data from the period 01-01-2024 to 30-04-2025 and then utilizing the models such as ARCH, GARCH (1,1), EGARCH, TGARCH and Unit Root tests. The results shows that returns of the ETF have constant irregularities, volatility clustering and persistence. This study also shows a significant heteroskedastic behaviour in ETF returns, thus emphasizing the need of advanced modelling by the policymakers and investors for effective risk management and investment strategy.

Keywords: ETFs, Volatility, ARCH, GARCH, Unit Root

CHAPTER – 1

INTRODUCTION

1.1 Introduction

Financial markets have undergone a revolution thanks to exchange-traded funds (ETFs), which give investors inexpensive, diversified, and liquid exposure to a range of asset classes. ETFs have had exponential growth since their launch in the early 1990s, and by 2024, their global assets under management (AUM) will have surpassed \$10 trillion (Joshi & Dash, 2024). Due to their quick growth, investment practices have changed, moving away from active management and toward passive management (Polisetty & Manda, 2019; Fichtner et al., 2017). But this expansion has also generated a great deal of discussion about how they affect systemic stability, price efficiency, and market volatility.

ETFs are frequently commended for improving liquidity and reducing transaction costs, but there are still worries that they may exaggerate market swings, especially in times of stress (Clements, 2020; Hejazi et al., 2015). While some studies assert that ETFs stabilize markets by absorbing shocks, others argue that they inject noise into asset prices through arbitrage processes (Moussawi et al., 2015) (Lin, 2024). The contrast between these viewpoints highlights how complicated ETF-induced volatility is, calling for a more thorough examination of its origins, trends, and wider market ramifications.

By reviewing the literature, identifying the main factors influencing ETF-related volatility, and evaluating the implications for financial markets, this dissertation aims to add to the current conversation. Regional differences will also be examined, especially in developing nations like India, where the use of ETFs is still relatively low despite their promising growth prospects (Tarasov, 2016; Polisetty & Manda, 2019).

1.2 Background and Evolution of ETFs:

With its introduction in 1993, ETFs launched with the launch of the SPDR S&P 500 ETF (SPY), designed to track the performance of the S&P 500 index. Institutional and ordinary investors soon embraced their structure, which combines the diversification of mutual funds with the tradability of equities (Madhavan, 2014). ETFs, as opposed to conventional mutual funds, trade intraday on exchanges, enabling investors to react quickly to changes in the market. They are a desirable substitute for actively managed funds due to their passive management approach, which also lowers expenses (Barber & Odean, 2000).

Financial markets have seen substantial structural changes in tandem with the growth of ETFs. With a combined AUM of billions, the "Big Three" asset managers—BlackRock, Vanguard, and State Street—now control the majority of the ETF market (Fichtner et al., 2017). Their involvement in active ownership and corporate governance raises worries about market concentration and systemic hazards, demonstrating their power beyond passive investing (Robertson & Lund, 2023). Furthermore, additional aspects of volatility have been brought

about by the growth of specialty ETFs, such as leveraged, inverse, and theme funds (Hejazi et al., 2015).

1.3 The Rise of ETFs: Promises and Paradoxes

ETFs were first hailed for their ability to improve the efficiency of the financial markets. Their structure promised to level the playing field for both institutional and retail investors by providing cost-effectiveness, intraday liquidity, and diversified exposure. ETFs typically track their underlying indexes well, sustaining significant comovement despite intraday pricing discrepancies, according to early research, including Aber et al. (2009).

But as ETFs gained traction, their impact on the larger financial system became more nuanced and contentious. While ETFs help with price alignment and arbitrage, Madhavan (2014) highlighted in his analysis of ETF trading patterns that they can also bring nonlinear dynamics that traditional Efficient Market Hypothesis models are unable to account for. Many academics have revisited and revised earlier optimistic assumptions regarding the effects of ETFs as a result of the appearance of phenomena like volatility clustering, tracking errors, and price distortions.

1.4 ETFs and Market Volatility: Divergent Scholarly Perspectives

The relationship between ETFs and volatility is one of the most hotly contested aspects of their market influence. With empirical results that both support and cast doubt on the role of ETFs in disrupting markets, the academic community is still split. On the one hand, Moussawi et al. (2015) provide strong proof that ETFs increase stock volatility. By using econometric techniques such as OLS and Regression Discontinuity Design, they discover that stocks that are owned by more ETFs have more price swings. They explain this by saying that the arbitrage mechanism, which is necessary to maintain ETF prices in line with Net Asset Values (NAVs), unintentionally magnifies transient price fluctuations.

Likewise, ETFs are criticized by Clements (2020) for encouraging herd mentality. ETFs' ease of trading promotes coordinated movements among various investor groups, amplifying asset price upswings and downswings. Because they are in line with the more general worries about market fragility in a setting where passive methods are becoming more and more prevalent, Clements' behavioural insights are essential.

On the other hand, some researchers adopt a more moderate stance. According to Lin (2024), for instance, ETFs offer liquidity shock absorption, particularly during volatile times, even though they are linked to a higher concentration of ownership. Depending on the overall state of the market, his modeling of the ETF mechanism implies a dual role that can be either stabilizing or destabilizing. Similar to this, Lynch et al. (2021) contend that ETFs primarily follow rather than drive the values of underlying assets based on high-frequency data. These opposing viewpoints imply that the relationship between ETFs and volatility varies

depending on the situation, which is a fundamental concept that informs my research topics and approach.

1.5 Behavioural Dynamics and Investor Implications

To understand the wider effects of ETFs on market volatility, one must have a solid understanding of investor behaviour. Barber and Odean (2000) offer fundamental understanding of the negative consequences of excessive trading and investor overconfidence. They discover that people frequently overestimate their trading prowess, which raises transaction costs and reduces net returns. This behavioural bias is especially pertinent to ETFs since its liquidity and real-time trading capabilities may entice investors to trade frequently, escalating the volatility patterns that are already present in ETF structures.

This relationship between ETF mechanics and investor psychology highlights a crucial aspect of market volatility that goes beyond structural characteristics. It emphasizes the necessity of looking at how the behavioural traits of various investor classes—algorithmic, institutional, and retail—interact with the ecology of the ETF market to affect volatility dynamics.

1.6 Passive Investing: Stability or Distortion?

Whether passive investing, as exemplified by ETFs, promotes or detracts from market efficiency is a major topic of discussion in the literature. Using the Grossman-Stiglitz paradox as support, Coles et al. (2022) contend that although passive investment lessens individual incentives for information gathering (fewer analyst reports, fewer views of EDGAR documents, etc.), active investors' equilibrium adjustments guarantee price efficiency. According to them, passive investing does not always result in a decline in the quality of market knowledge.

The stock prices of large-cap companies are disproportionately inflated by passive investment inflows, which distort asset values and increase idiosyncratic volatility and inefficient capital allocation, according to Vayanos et al. (2024). Their findings are especially pertinent to large indices, such as the S&P 500, where firms see greater volatility and post-inflow returns. The dangers of misallocation that result could have a significant impact on financial stability, resource distribution, and economic growth.

The idea of "Activeness" in ETFs is introduced by O'Hara et al. (2021), further confounding the conventional narrative. They contend that most ETFs have active management features that have a big impact on market share, costs, and price discovery; most are not just passive in form or function. Academic classifications and regulatory frameworks that rely on a distinct active-passive dichotomy are challenged by the increasing "hybrid" character of ETFs.

1.7 Special Contexts: Emerging Markets, Leveraged ETFs, and Corporate Governance

The majority of the literature now in publication is on developed countries, such as the US and Europe, while my research focuses especially on emerging markets, where ETF dynamics might vary greatly. According to studies like Tarassov (2016) and Polisetty and Manda (2019), ETF adoption is slower, investor behavior is more cautious, and volatility patterns differ from those seen in industrialized nations in markets like India. Additionally, Khanapuri's analysis of GoldBees and NiftyBees using GARCH and ARCH models reveals strong volatility clustering, indicating that any generalizable model of ETF-induced volatility must take into account market-specific and asset-specific characteristics.

Another level of complication is added by leveraged ETFs (LETFs). Hejazi et al. (2015) show that although LETFs have the ability to increase intraday volatility, their tiny market size limits their overall systemic risk. However, my dissertation delves deeper into the possibility that the behavioral attraction of LETFs throughout boom and bust cycles could have significant psychological and local market effects. Concern over corporate governance is also on the rise. The concentration of voting power in a small number of ETF sponsors (the "Big Three") has changed the dynamics of shareholders, as Fichtner et al. (2017) note. Robertson and Lund (2023), however, warn against oversimplifications by demonstrating how active and passive distinctions can get hazy in real-world situations.

1.8 The Way Forward: Gaps and Emerging Trends

Finally, Joshi and Dash (2024) offer a thorough bibliometric analysis, emphasizing recent research trends and knowledge gaps. While highlighting the United States' dominance in ETF research, they also highlight the increasing importance of subjects like artificial intelligence (AI) applications, ESG (Environmental, Social, and Governance) investing, and the use of ETFs in emerging markets. Their suggestions highlight the need for future research to examine volatility and systemic hazards associated with ETFs from a global, multifaceted standpoint.

ETFs' transformation from specialized goods to powerful market players has been nothing short of revolutionary. However, as demonstrated in this introduction, the effects of ETFs on market volatility are complex, ever-changing, and frequently incongruous. ETFs can introduce noise and distort prices, according to some studies, but they can also stabilize markets and increase market efficiency. These results are shaped by a combination of global trends, regulatory frameworks, investor behavior, and structural market shifts.

CHAPTER – 2

LITERATURE REVIEW

2.1 Literature Review:

To understand different viewpoints on this topic of volatility in ETFs, by various scholars and researchers, a review of relevant literature and scholarly articles is presented that spans across both theoretical and empirical studies to dive deep into this topic.

Moussawi et al. (2015) examines the influence of ETF on stock market volatility by drawing noise trends, resulting in price instability through arbitrage mechanism. By applying OLS regressions and RDD (Regression Discontinuity Design), they draw a link between holding an ETF and a rise in price instability. The outcome indicates that passive investing does not enhance market efficiency but brings a new layer of noise.

Barber and Odean (2000) analyses individual investors and their behaviour and how overconfidence and excessive trading cause bad net returns due to high transaction costs. The paper also shows how trading costs such as commission, reduce investor profits and highlight biasness in investment decisions.

Coles et al. (2022) examine the effect of passive investment on market efficiency through a Grossman-Stiglitz model and Russell Index reconstitutions. They conclude that passive investment decreases information production (e.g., less analyst coverage, EDGAR views), but price efficiency is unaffected because active investors reallocate to keep the market informative. This implies that index investment does not adversely affect market efficiency due to equilibrium mechanisms that ensure informational quality.

Lin (2024) examines the debate of how ETFs' affect stock volatility, does it contribute to market volatility or if they help stabilize the market. Lin proposes that these both mechanisms coexist, and how stocks that are owned more by ETFs' exhibits more volatility, compared to stocks that are traded more in ETF market which absorbs the fluctuation and keep prices more stable.

Tarassov (2016) provides an overview of the academic literature on ETFs, focussing on their development, working mechanism, and impact in the market. The article also points out notable research gaps, including investor choices and behaviour and low adoption of ETFs in the emerging markets, and suggests a new framework of classification to be used for future research.

Polisetty and Manda (2019) examines the growth of ETFs in India, focusing on market dynamics, regulations, and the government's use of ETFs for disinvestment, demonstrating a move away from active to passive investing. They draw attention to Indian investor's inclination for active management and their lack of knowledge about ETFs.

Hejazi et al. (2015) investigates the impact of leveraged ETFs on end-of-day volatility in order to address concerns expressed by regulators and market participants regarding the possibility that these funds could increase volatility through hedging operations. Some claim that these funds are too small to significantly affect volatility. By objectively analysing these claims using financial crisis data from 2006 to 2011, the study contributes to the conversation.

Vayanos et al. (2024) investigates how passive investing affects asset prices and finds that stock prices of huge, frequently overpriced companies are disproportionately raised by inflows into passive funds. The authors show that, especially for S&P 500 companies, which have the highest returns and volatility after inflow, these flows increase idiosyncratic volatility and deter price corrections. According to the findings, passive investment lowers financing costs for large firms and skews the distribution of firm sizes, which may result in capital misallocation.

O'Hara et al. (2021) challenge the traditional view that ETFs are purely passive investment vehicles. They demonstrate that the majority of ETFs are either active in function or active in form. ETFs are becoming more active, which has significant ramifications for fees, market share, and price discovery, according to their new Activeness Index. Their research expands on current theories regarding active versus passive investment and contends that the emergence of exchange-traded funds (ETFs) improves rather than detracts from market efficiency.

Lynch et al. (2021) - Although previous study indicated that ETFs enhance price discovery, more recent studies have raised concerns about ETFs using arbitrage to introduce noise into underlying stocks. A large portion of the literature, meanwhile, makes this assumption without conducting intraday testing. In order to fill that vacuum, this study suggests that ETFs primarily follow their constituents rather than disrupting them, finding no evidence that ETF trading affects underlying pricing.

Fichtner et al. (2017) examines how the "Big Three" (BlackRock, Vanguard and State Street) have risen to prominence as passive investors and owners of US firms. It examines their convert impact over corporate governance and centralised voting practices. The report also emphasises how new financial dangers are brought forth by their emergence.

Robertson and Lund (2023) examines the misconceptions caused by the frequent abbreviation that links index funds, passive investment, and the Big Three (BlackRock, Vanguard, and State Street). It demonstrates that the Big Three handle large active assets in addition to their completely passive ones. The study draws attention to the distinctions between the Big Three and the underappreciated contribution of other titans, such as Fidelity. The index funds themselves, it is said, are very diversified and not really "passive". Lastly, it talks about how these findings affect corporate governance and policymaking.

Madhavan (2014) investigates nonlinearity in Indian ETFs. In order to challenge the Efficient Market Hypothesis (EMH), the literature on market efficiency has moved away from linear models and toward nonlinear serial dependence, chaos, and long memory behaviours. Asset return nonlinearity points to increased risk and possible predictability. Research conducted in both established and developing markets has found the evidence of nonlinearity, which is frequently seasonal rather than continuous.

Khanapuri (2019) looks at how the GARCH and ARCH models are used to investigate volatility in Indian ETFs, particularly the commodity-based GoldBees and the equity-based NiftyBees.

The findings support GARCH (1,1) modelling, which successfully captures volatility clustering, by confirming large ARCH effects. The underlying assets (gold and the Nifty) have a major impact on returns, and the model shows good forecasting accuracy. The results support the applicability of GARCH models for ETF volatility analysis in emerging markets and are consistent with previous studies.

According to **Clements (2020)**, ETFs have the potential to increase market volatility by promoting herd mentality and skewing price discovery. It draws attention to the dangers of high-frequency trading, intermediary concentration, and passive investing, all of which have the potential to cause liquidity crises. While recognizing the advantages of ETFs, the report urges regulatory changes to address these systemic risks.

Joshi and Dash (2024), in this study, review the future of passive investments and exchange-traded funds (ETFs), highlighting significant subjects like volatility, liquidity, and tracking errors as well as important trends, well-known writers, and prestigious publications. The study emphasizes how the USA leads the world in ETF research as well as new areas like AI applications and ESG investment. It also identifies knowledge gaps and recommends future courses of inquiry, especially for developing nations.

Aber et al. (2009) compares four iSharesTM exchange-traded funds (ETFs) to traditional index mutual funds that track the same standards in order to assess the price volatility and tracking performance of the ETFs. To assess performance, the study uses metrics like tracking errors, daily returns, and premium/discount positions. The main conclusions are that, although tracking errors are different between the two fund types, ETFs typically trade at premiums, show notable intraday price swings, and generally demonstrate average cross correlation with benchmarks that is comparable to that of mutual funds.

Brown and Pomerantz (2024) addresses gaps in literature and research studies regarding ETFs. To find the extent to which ETF assets are utilized for trading instead of long-term passive methods, the study examines ETF measures such as fund size, fee ratios, and turnover rates. It examines how options trading affects the turnover of exchange-traded funds (ETFs) and queries the actual percentage of assets in the investment sector that are managed passively.

CHAPTER – 3

DATA AND METHODOLOGY

3.1 Objectives of the research

1. To examine the behaviour of ETF volatility.
2. Modelling ETF using ARCH and GARCH model.

3.2 Research Framework

3.2.1 Need of study

A lot of fluctuations took place in the ETF's closing prices on day-to-day basis.

3.2.2 Sample Frame and Sample Size

We took daily statistics of particular ETF called UTI Nifty IT ETF (NIFITETF) from January 1st 2024 to April 30th 2025 to evaluate volatility of these cryptocurrencies.

3.2.3 Analysis Techniques

The ARCH and GARCH methodology are applied for the process of volatility analysis.

3.2.4 Tools used for analysis

The graph of the ETF (Nifty IT ETF) closing prices was plotted with the assistance of Microsoft Excel and all of the other analysis required for the ARCH and GARCH model was carried out in E-views.

3.3 Implications of the study

By evaluating the historical performance of the volatility of closing prices, we can acquire insight into the future conditions. Calculating the returns is essential for determining the level of risk involved. This determination is essential for investors, corporations and traders in making required decisions. In addition, both students and academics conducting more research will find this helpful.

CHAPTER – 4

EMPIRICAL ANALYSIS

4.1 Empirical Analyses

We have taken ETF's daily closing prices data from 1st January, 2024 to 30th April, 2025 to study ETF returns volatility as seen in Figure 1. Here we can see there is fluctuations in the graph indicating volatility in closing price of the ETF.

Figure 1: Graph of closing price of ETF

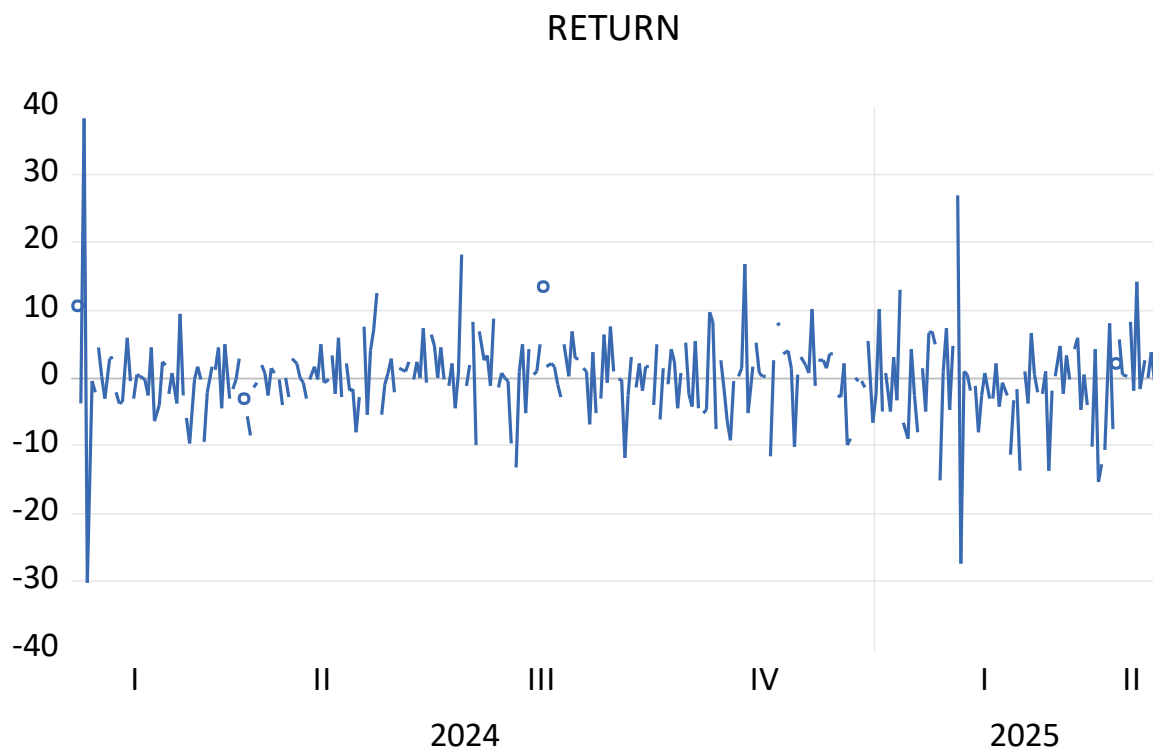


(Authors Calculation)

The data exhibits notable oscillations, encompassing both positive and negative market trends. The second and third quarters of 2024 saw an increase, with the closing price peaking at about 460. In the first quarter of 2025, however, the trend abruptly reverted, signalling a period of downturn and elevated volatility. There are indications of a possible recovery in the latest data. All things considered, this graph demonstrates the NIFTYETF's dynamic behaviour and lays the groundwork for additional research into its volatility and return characteristics.

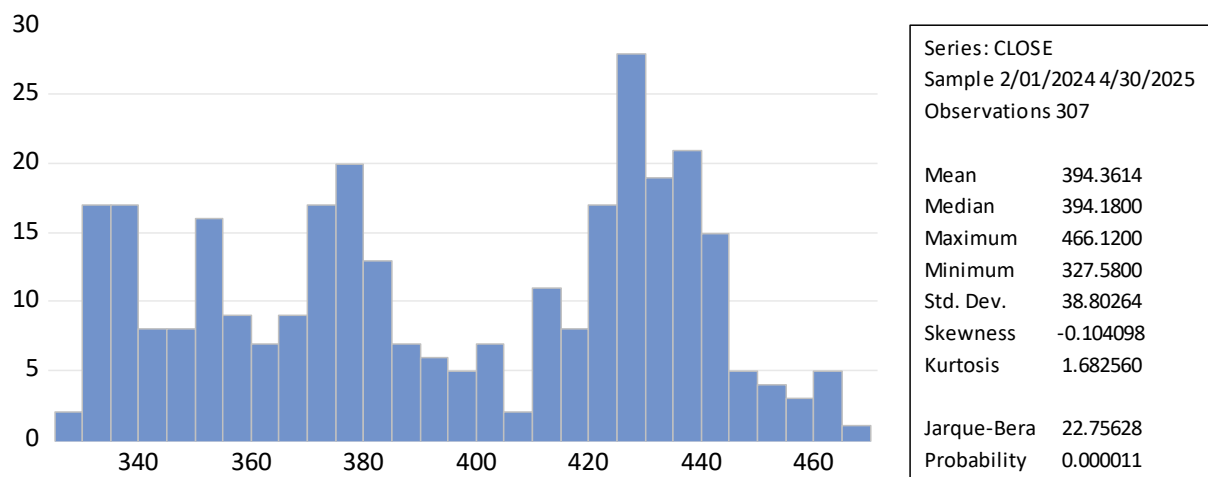
The graph below in figure 2 shows return of the ETF. It's nothing but the first difference of the closing price values and span the same time frame.

Figure 2: Graph of ETF returns



(Authors Calculation)

Figure 3: Jarque Bera Probability data of NIFITETF



(Authors Calculation)

So, from the above table we can infer that:

Median value is 394.1800, indicating the central value of the data series.

Maximum value is 466.1200 and Minimum value is 327.5800 which shows highest and lowest value in the series.

Deviation from the average value of the series is 38.80264.

Value of skewness is -0.104098, signifying that data may be slightly asymmetric.

Value of kurtosis is 1.682560, denoting that the distribution produces fewer and less extreme outliers as compared to the normal distribution.

Jarque-Bera is a test for ensuring the normal distribution. This test is done before any other test to ensure everything is okay. Returns do not follow a normal distribution, rejecting zero skewness and kurtosis. Now by considering the change in daily logarithmic returns:

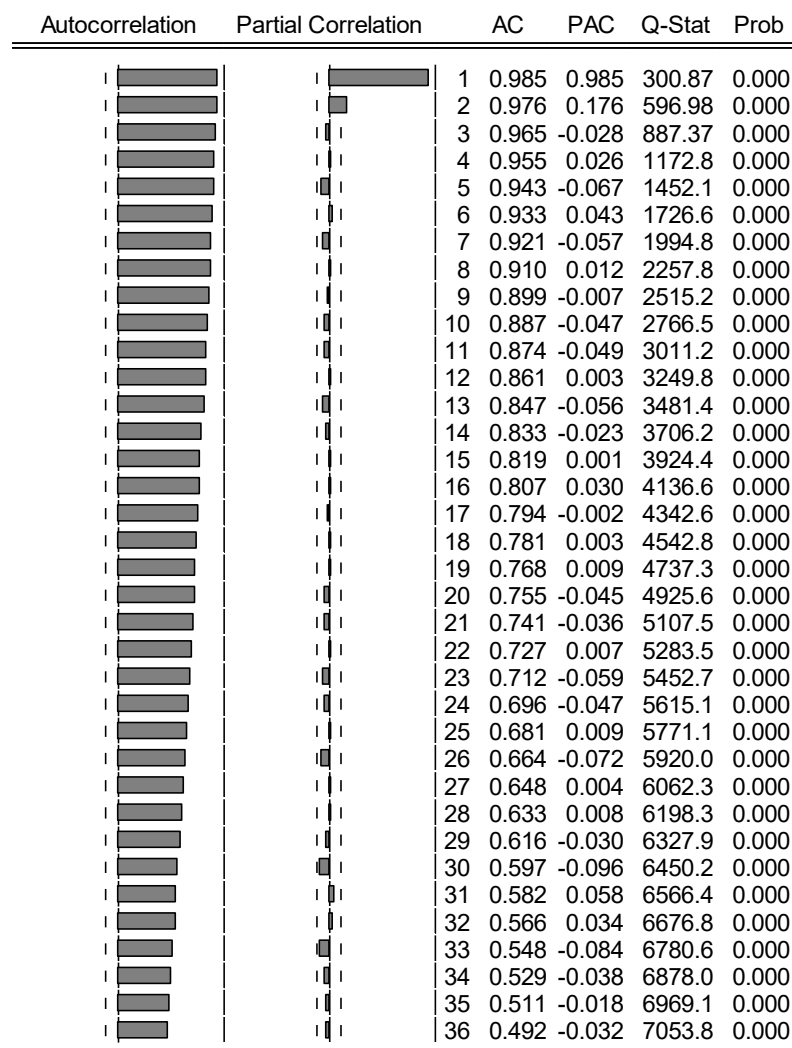
Taking 'mt' as the closing price at time t , then the series can be altered as:

$$\begin{aligned} P_t &= \log (m_t / m_{t-1}) \\ &= \log (m_t) - \log (m_{t-1}) \end{aligned}$$

Here, P_t represents daily closing price returns.

Standard deviation helps in reflecting the normal volatility. It is undetected in this financial series. We will try to model this volatility using well-known and frequently used Auto Regressive Conditional Heteroscedasticity (ARCH) and Generalised Auto Regressive Conditional Heteroscedasticity (GARCH) in order to capture both symmetric and asymmetric effects.

Figure 4: Correlogram of Standardized Squared Residuals of NIFITETF



(Authors Calculation)

From the figure above we can see, that across several lags the autocorrelation (AC) and partial autocorrelation (PAC) values stay noticeably high, and the Q-statistics are sizable, with p-values for each of the 36 lags equal to 0.000. This suggests that there is volatility clustering in the series and clearly shows serial correlation in the squared residuals.

These results support the existence of conditional heteroscedasticity by indicating that the variance of the residuals is not constant across time. Consequently, it allows for the proper modelling and forecasting of the volatility in NIFTYETF returns using the ARCH and GARCH models.

Table 1 & 2: Testing for Stationarity using Augmented Dickey-Fuller (ADF) Test

	t-Statistics	Probability
ADF test statistics	-1.421202	0.5721
Test critical value at 1% level	-3.451561	
5% level	-2.870774	
10% level	-2.571761	

(Author Calculation)

This test is used to determine whether the closing price of the NIFITETF is stationary or contains a unit root. As the p-value is greater than 0.05, we fail to reject the null hypothesis which means it is non stationary. But to proceed with ARCH/GARCH modelling it is important to achieve stationarity, so we will test for unit root in 1st difference rather than level like we did above.

	t-Statistics	Probability
ADF test statistics	-20.99783	0.0
Test critical value at 1% level	-3.451632	
5% level	-2.870805	
10% level	-2.571777	

(Author Calculation)

The t-statistics is less than all the critical values at 1%, 5% and 10% levels and p-value is less than 0.05 (0.0000). This mean we can reject null hypothesis and conclude that the first differenced series is stationary.

Table 3: Estimating an ARMA model

In this step, we will estimate the ARMA model by the methodology of Conditional Least Squares method, more specifically by using the Gauss-Newton/Marquardt steps. This technique of estimation reduced the sum of squared residuals by assuming that the error terms are conditionally normally distributed.

Variable	Coefficient	Standard Error	t-Statistics	Probability
AR (1)	-0.793867	0.074335	-10.67958	0.0000
MA (1)	0.782057	0.084879	9.213826	0.0000

(Authors Calculation)

The coefficient for the variable AR (1) is -0.79, which means current value is being negatively influenced by the last values or previous values.

The t-statistics and p-value indicate this is highly significant.

The value of coefficient for MA (1) is 0.78 which mean the current error is influenced positively by previous shocks.

Figure 5: Correlogram of Residuals

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1 -0.077	-0.077	1.8274	
		2 -0.058	-0.064	2.8702	
		3 0.104	0.096	6.2536	0.012
		4 -0.018	-0.006	6.3534	0.042
		5 -0.072	-0.064	7.9929	0.046
		6 0.041	0.020	8.5216	0.074
		7 -0.036	-0.037	8.9290	0.112
		8 0.002	0.013	8.9303	0.178
		9 0.031	0.022	9.2392	0.236
		10 0.049	0.058	9.9953	0.265
		11 -0.040	-0.029	10.512	0.311
		12 0.060	0.052	11.676	0.307
		13 0.009	0.008	11.702	0.386
		14 -0.036	-0.020	12.122	0.436
		15 -0.013	-0.022	12.176	0.513
		16 0.064	0.055	13.523	0.486
		17 -0.003	0.022	13.526	0.562
		18 -0.000	0.004	13.526	0.634
		19 0.058	0.048	14.633	0.622
		20 0.051	0.057	15.478	0.629
		21 -0.051	-0.032	16.328	0.635
		22 0.110	0.095	20.358	0.436
		23 -0.019	-0.009	20.479	0.491
		24 -0.041	-0.018	21.046	0.518
		25 0.070	0.048	22.687	0.479
		26 -0.022	-0.020	22.844	0.529
		27 -0.043	-0.015	23.454	0.551
		28 0.045	0.009	24.141	0.568
		29 0.080	0.087	26.332	0.500
		30 -0.092	-0.073	29.230	0.401
		31 0.027	0.012	29.471	0.441
		32 0.084	0.052	31.880	0.373
		33 -0.011	0.031	31.923	0.420
		34 0.004	0.006	31.929	0.470
		35 0.037	0.010	32.416	0.496
		36 -0.070	-0.053	34.121	0.462

(Author Calculation)

4.2 ARCH LM Test

Regressing the squared residuals on a constant and p lags of the residuals is usually involved in this test, where selection of p is according to which lag minimises an information requirement.

Null Hypothesis: The residuals show no signs of ARCH effects.

Alternative Hypothesis: The residuals contain ARCH effects.

Consequently, the residuals suffer from ARCH if the p -value is less than 0.05.

Table 4: Heteroskedasticity Test for ARCH

H0: There are no ARCH effects up to the specified lag order.

H1: There are ARCH effects up to the specified lag order.

If $p < 0.05$, we reject H0 and confirm the existence of ARCH effects.

Variables	Coefficient	Standard Error	t-Statistics	Probability
C	22.00712	4.933828	4.460456	0.0000
RESID ^2 (-1)	0.365504	0.053419	6.842280	0.0000

(Authors Calculations)

The result shows that there exist ARCH effects, which makes it appropriate for further study.

That's because F statistics is 46.81679 with p value of 0.0000.

The Obs*R-squared statistics is 40.80166 and p value 0.0000.

As p values are less than 0.05, we reject null hypothesis at 5% significance level.

Figure 6: Correlogram Squared Residuals

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.365	0.365	41.115	0.000
		2	0.013	-0.139	41.168	0.000
		3	-0.007	0.045	41.183	0.000
		4	-0.028	-0.047	41.430	0.000
		5	0.048	0.089	42.163	0.000
		6	0.064	0.009	43.453	0.000
		7	-0.028	-0.060	43.705	0.000
		8	0.000	0.047	43.705	0.000
		9	0.010	-0.011	43.737	0.000
		10	-0.035	-0.038	44.126	0.000
		11	0.020	0.047	44.248	0.000
		12	-0.042	-0.083	44.816	0.000
		13	-0.065	-0.009	46.184	0.000
		14	0.047	0.075	46.894	0.000
		15	0.048	0.001	47.647	0.000
		16	-0.001	-0.018	47.648	0.000
		17	0.070	0.089	49.246	0.000
		18	-0.008	-0.065	49.270	0.000
		19	-0.020	0.017	49.399	0.000
		20	-0.002	-0.026	49.401	0.000
		21	-0.059	-0.042	50.553	0.000
		22	-0.012	0.024	50.597	0.000
		23	0.038	0.018	51.088	0.001
		24	-0.022	-0.034	51.255	0.001
		25	0.057	0.089	52.358	0.001
		26	0.033	-0.035	52.728	0.001
		27	-0.040	-0.008	53.273	0.002
		28	-0.035	-0.041	53.686	0.002
		29	-0.018	0.017	53.793	0.003
		30	-0.005	0.004	53.801	0.005
		31	0.032	0.002	54.143	0.006
		32	-0.030	-0.049	54.441	0.008
		33	-0.021	0.035	54.596	0.010
		34	0.016	-0.017	54.680	0.014
		35	-0.057	-0.046	55.812	0.014
		36	-0.042	-0.002	56.439	0.016

(Author Calculation)

The figure 6 is a correlogram of squared residuals which is a test for volatility clustering, which checks heteroskedasticity. It shows autocorrelation and partial autocorrelation functions of the squared residuals from ARMA.

From the above figure we can see a strong initial autocorrelation, and also evidences of ARCH effects, justifying a GARCH model.

Q statistics (Ljung Box Test) is significant for each one of the lags in the table as $p < 0.05$, which means it reject null hypothesis of no autocorrelation.

Table 5: Estimating ARCH Model

With 'return' as our dependent variable, we will be estimating the ARCH model by the methodology of ML – ARCH Normal Distribution more particularly Marquardt/Eviews legacy settings. The result thus generated were as follow

Variable	Coefficient	Standard Error	z-Statistics	Probability
AR (1)	0.661715	0.588122	1.125132	0.2605
MA (1)	-0.652650	0.595969	-1.095108	0.2735

Variance Equation

C	25.62050	1.947285	13.15703	0.0000
RESID (-1) ^ 2	0.310610	0.063524	4.889664	0.0000

(Authors Calculation)

0. Significance of Coefficients

ARCH Term (RESID (-1) ²): Coefficient = 0.3106 and its p-value = 0.0000 indicating it is significant & indicates that past squared shocks (volatility clusters) significantly affect current volatility.

2. As the model is ARCH (1) (no GARCH term), persistence is driven solely by α . $\alpha = 0.3106$: Moderate short-term volatility persistence (no long-term memory as β is absent).

3. AR/MA Terms (Mean Equation)

AR (1) and MA (1): Coefficients are insignificant (p-values > 0.26), suggesting no strong linear dependence in returns.

4. Key Takeaways: Volatility is driven by ARCH effects (past shocks), but lacks GARCH-driven persistence. Model may benefit from extending to GARCH (to capture long-term volatility memory). Low R-squared (0.0002) implies the model explains little return variation (typical for volatility models).

4.3 Generalized Autoregressive Conditional Heteroscedasticity Model

After looking at the data from our selected time frame for ETF, we will now be estimating GARCH (1,1) Model for checking volatility or volatility clustering because it outperforms ARCH Model in breaking non-negativity returns.

Table 6: Estimating GARCH Model

The GARCH or in this particular case GARCH (1,1) model is a statistical time series model which is used in financial markets and other fields where there is need to forecast and analyse volatility and where volatility clustering is evident. In this case we have used the methodology of ML ARCH – Normal distribution more specifically BFGS (Broyden Fletcher Goldfarb Shanno)/Marquardt steps. The following are the generated results:

Variables	Coefficients	Standard Error	z-Statistics	Probability
C	0.054500	0.314158	0.173479	0.8623
AR (1)	-0.058765	0.060515	-0.971077	0.3315

With variance equation being,

C	10.42642	2.802467	3.720442	0.0002
RESID (-1) ^2	0.078738	0.030602	2.572990	0.0101
GARCH (-1)	0.581958	0.094796	6.139032	0.0000

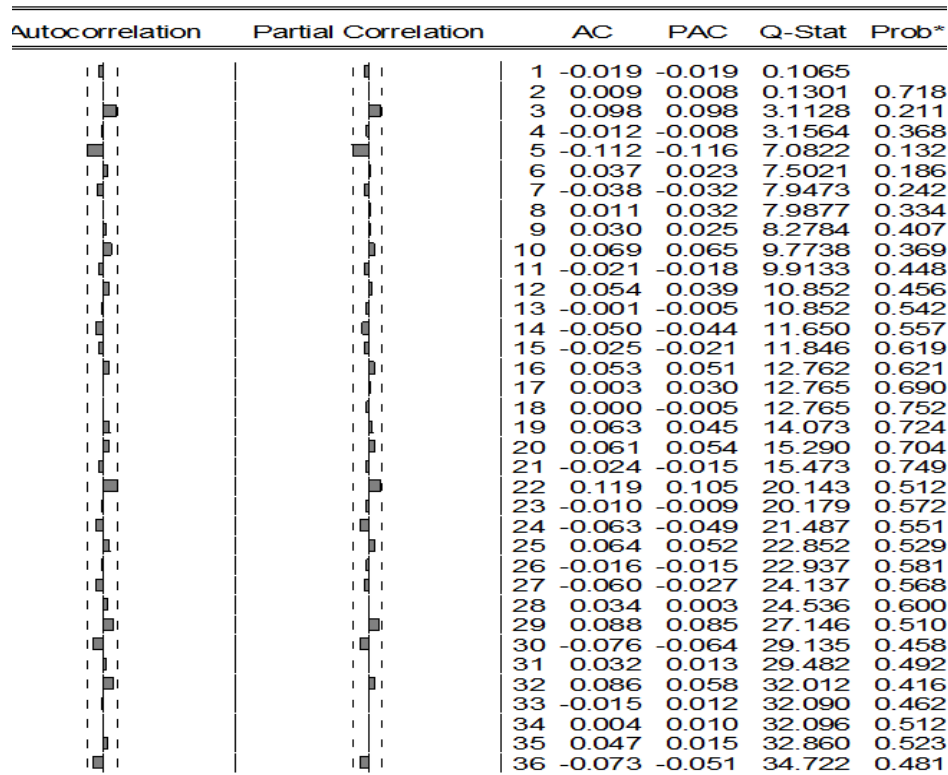
(Authors Calculation)

By looking at the probability values of C variable, which is 0.8623, we can observe that the data is non stationary.

Variance Equation for NIFITETF = $10.426 + 0.58 \varepsilon^2_{t-1} + 0.078 \phi^2_{t-1}$

The efficiency of fit of ARCH-GARCH is calculated using residuals. We will be doing five (5) types of Residual Diagnostics: Correlogram Q Statistics, Correlogram Squared Residuals, Histogram Normality Test, ARCH LM Test and Sign Bias Test.

Figure 7: Correlogram of Standardized Residual for GARCH (1,1)



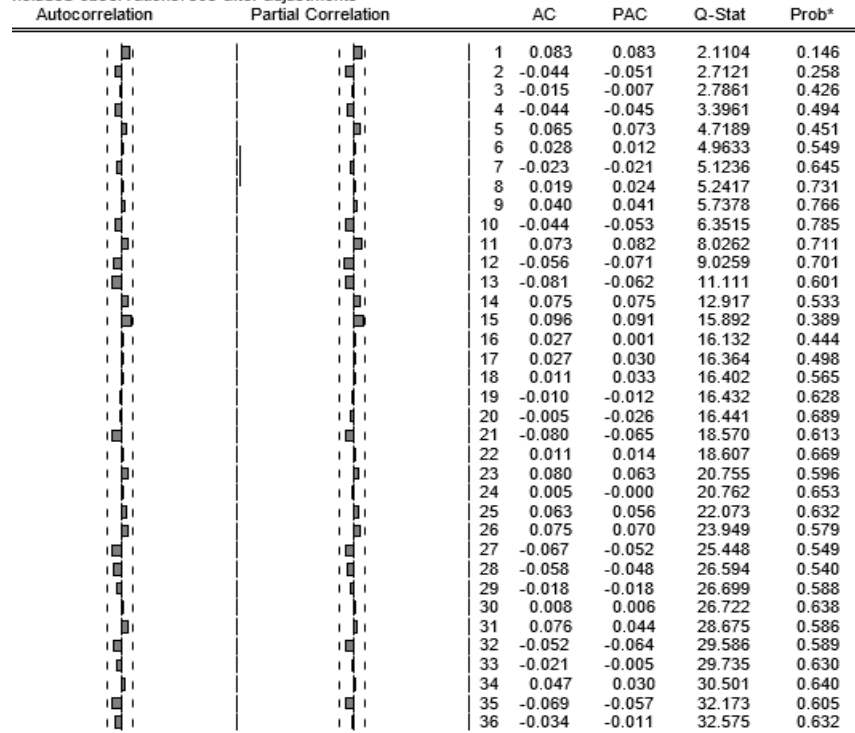
(Author Calculation)

Figure 8: Correlogram squared residual

Date: 05/15/25 Time: 16:35

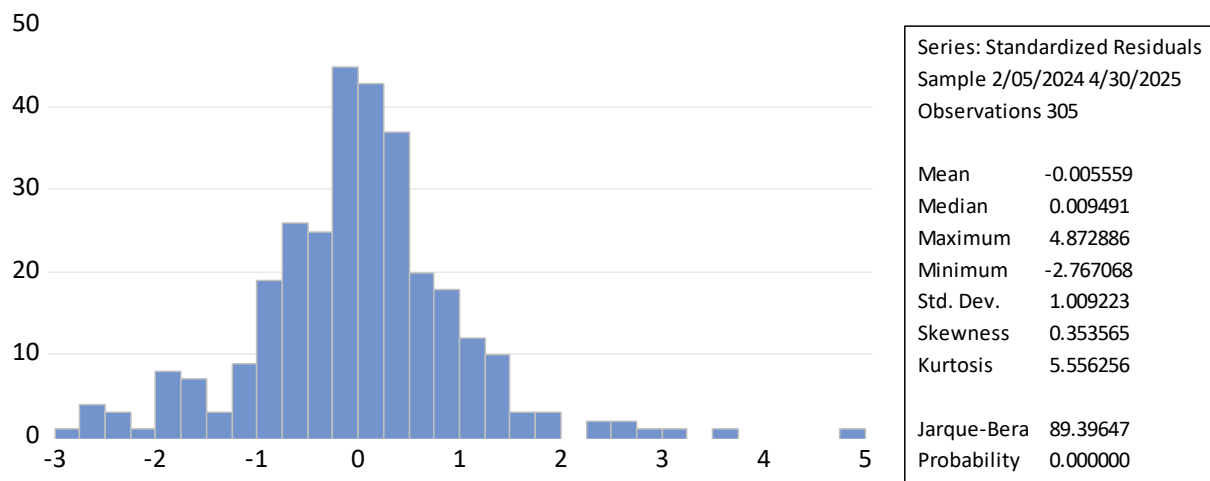
Sample (adjusted): 2/05/2024 4/30/2025

Included observations: 305 after adjustments



(Author Calculation)

Figure 9: Histogram Normality test



(Authors Calculation)

From the figure above, we can observe that:

Median Value is 0.009, indicating central value of the data.

Highest and lowest values are denoted by maximum and minimum, which are 4.87 and -2.76 respectively.

Standard Deviation is 1.009, which means deviation from the average value.

Value of skewness is 0.3535, signifying the symmetry or lack of it in the data

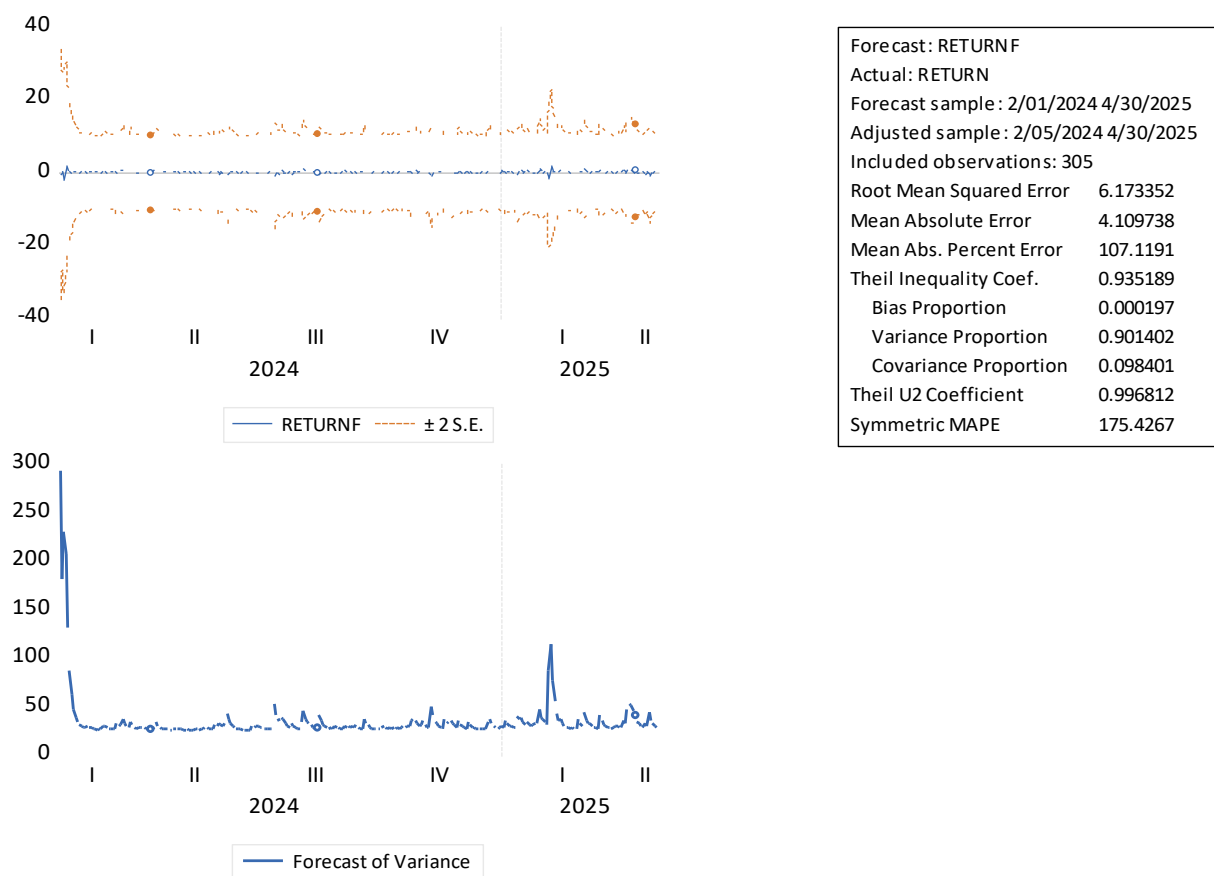
Table 7: Arch LM test

The ARCH LM test is a Lagrange Multiplier test for ARCH effects which is a statistical test which helps in identifying the presence of heteroskedasticity especially ARCH in time series residuals. The GARCH type model is suitable for modelling of volatility when ARCH effects are found. The following are the result generated:

Variables	Coefficient	Standard Error	t-Statistics	Probabilities
C	0.934074	0.137362	6.800100	0.0000
WGT_RESID ^2 (-1)	0.082833	0.057347	1.444420	0.1497

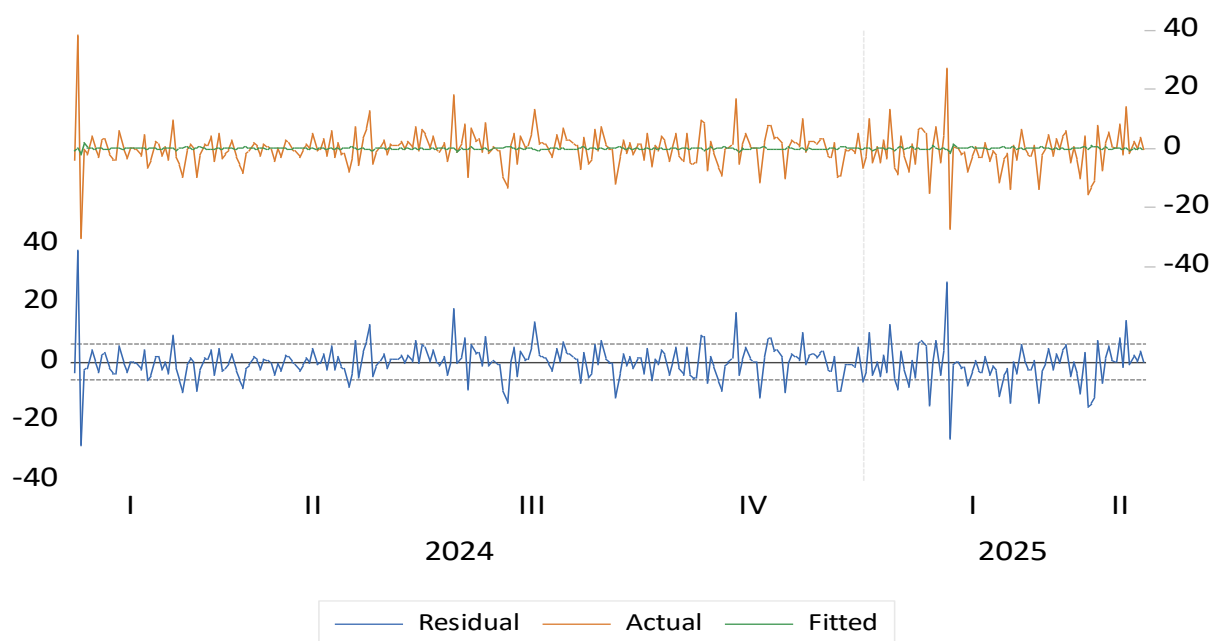
(Authors Calculation)

Figure 10: Forecasting using Static Forecasting Method



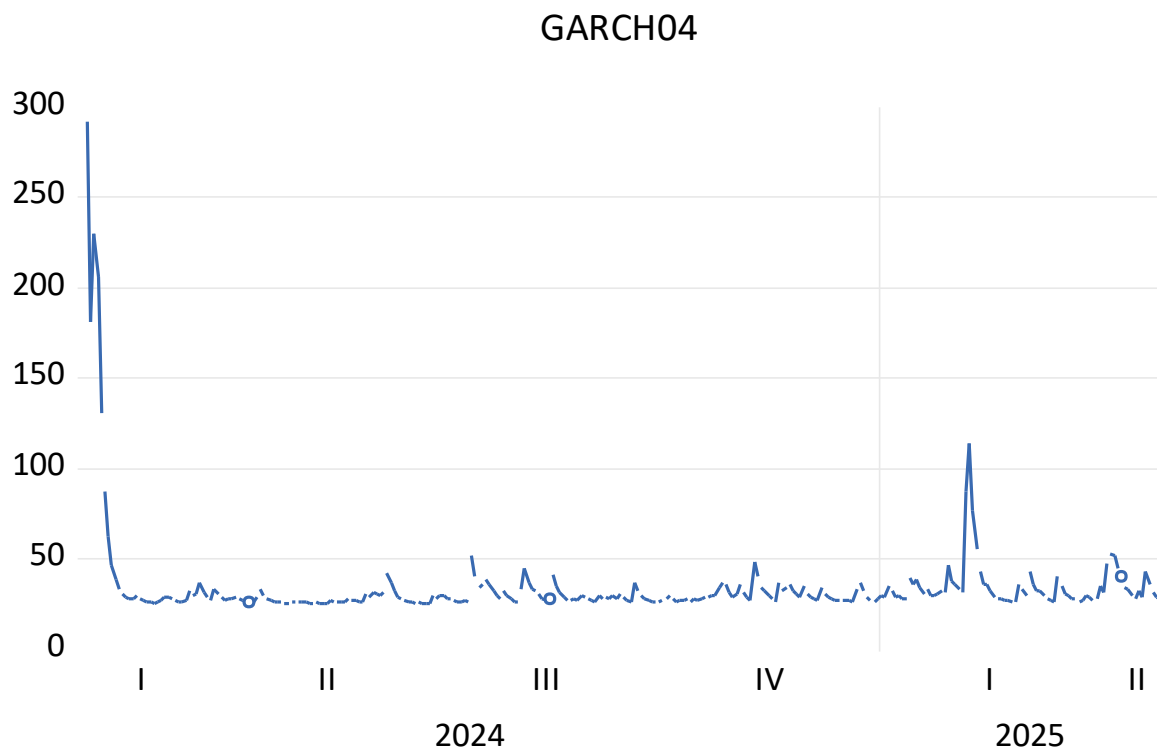
(Authors Calculations)

Figure 11: Residuals



(Authors Calculation)

Figure 12: GARCH Variance



(Authors Calculation)

The graph in figure 12 is a GARCH variance plot for our GARCH (1,1) model, in which the Y axis is variance, and X axis is the time span from the early 2024 to the starting of second quarter of the year 2025. This graph shows how volatility change through the time span and how it is uncertain. A common trend that can be observed is that there is a spike in early 2025 and 2024, with an extreme one in 2024 starting, that shows high volatility which can be caused because of sudden event that may have acted as shock in the data. As expected, variance didn't hit zero, which is typical of FARCH models.

4.4 The Leveraged Effect of Asymmetric GARCH Model

It is the relationship between historical gains and possible volatility in the future. Volatility usually goes up when price of an ETF decreases but decrease in price may be followed by increase in return. This makes a negative correlation between returns and volatility.

4.5 Estimate TARCH Model

The TARCH Model is an extension of GARCH model, in which the threshold order is taken as 1. It is also known as Threshold GARCH. In this model, the negative and positive shocks are allowed to have impact on volatility that is different.

The conditional variance equation in a TARCH(m,n) model is:

$$\sigma_t^2 = \omega + \sum(\alpha_i \varepsilon_{t-i}^2) + \sum(\beta_j \sigma_{t-j}^2) + \sum(\gamma_k \varepsilon_{t-k}^2 I_{t-k})$$

Where:

- $I_{t-k} = 1$ if $\varepsilon_{t-k} < 0$ (negative shock), and 0 otherwise
- γ_k captures the additional impact of negative shocks
- If $\gamma_k > 0$, negative shocks have a larger impact than positive shocks

Table 8:

Through methodology of ML ARCH Normal distribution test, following results were generated:

Variables	Coefficient	Standard Error	z-Statistics	Probability
C	0.150742	0.236577	1.674545	0.0001
AR (1)	-0.023877	0.045634	3.564577	0.1422

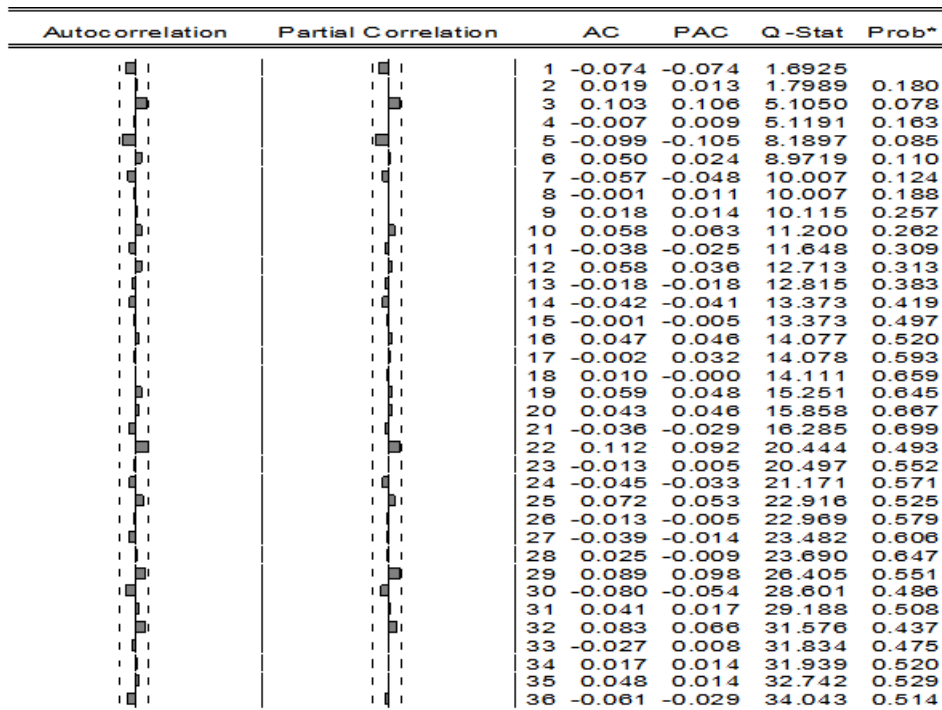
And the Variance equation being,

C	16.50206	0.654655	2.876868	0.0034
RESID (-1) ^2	0.020555	2.546758	1.465465	0.0177
RESID (-1) ^2*RESID (-1) <0	-0.084112	0.546578	0.754765	0.2457
GARCH (-1)	0.563554	1.654678	0.004446	0.0012

(Authors Calculation)

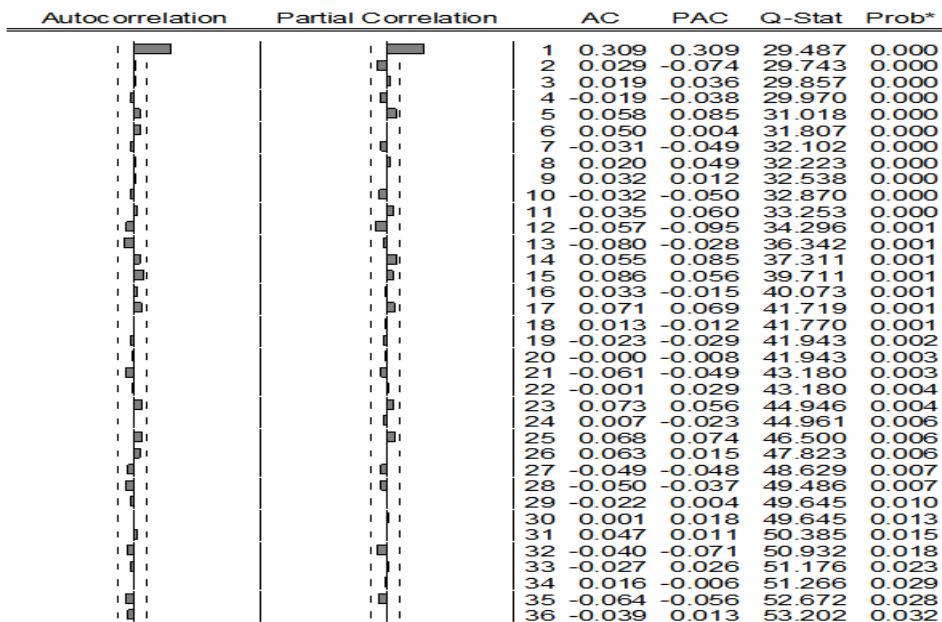
4.5.1 Residual Diagnostics

Figure 13: Correlogram Q Statistics



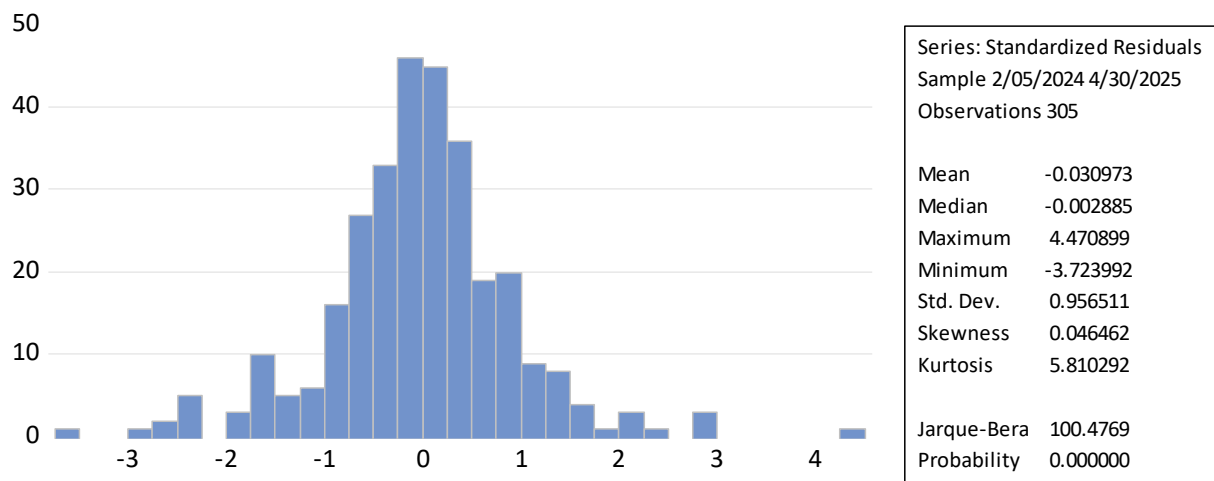
(Authors Calculation)

Figure 14: Correlogram squared residual



(Authors Calculation)

Figure 15: Histogram normality test



(Authors Calculation)

In figure 15, the values that are given are:

Median value is equal to -0.002 which means the central value of the data.

Maximum and Minimum are 4.47 and -3.72 respectively.

Deviation from the average value is 0.95

Value of skewness, which is checked to see the symmetry of data is 0.04.

Table 9: Arch LM test

Arch LM Test through methodology of least squares:

Variables	Coefficients	Standard Error	t-Statistics	Probability
C	0.632148	0.120477	5.247024	0.0000
WGT_RESID ^2 (-1)	0.309616	0.054718	5.658415	0.0000

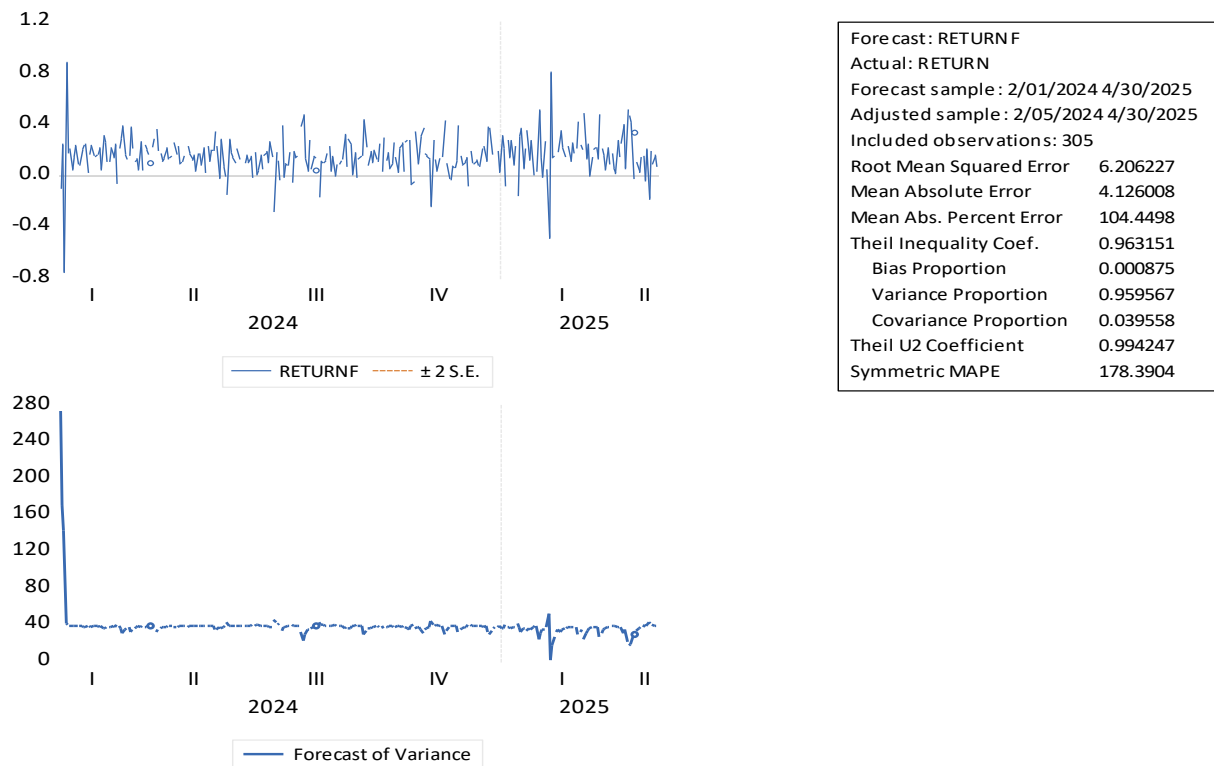
(Authors Calculation)

ARCH LM Test is used to check ARCH effects in residuals of time series model.

In table 9, F statistics is 32.017, and prob(F) is 0.0000. as p value < 0.05, there is no ARCH effect as even F statistics is high. This means, in the residuals, there is no heteroskedasticity.

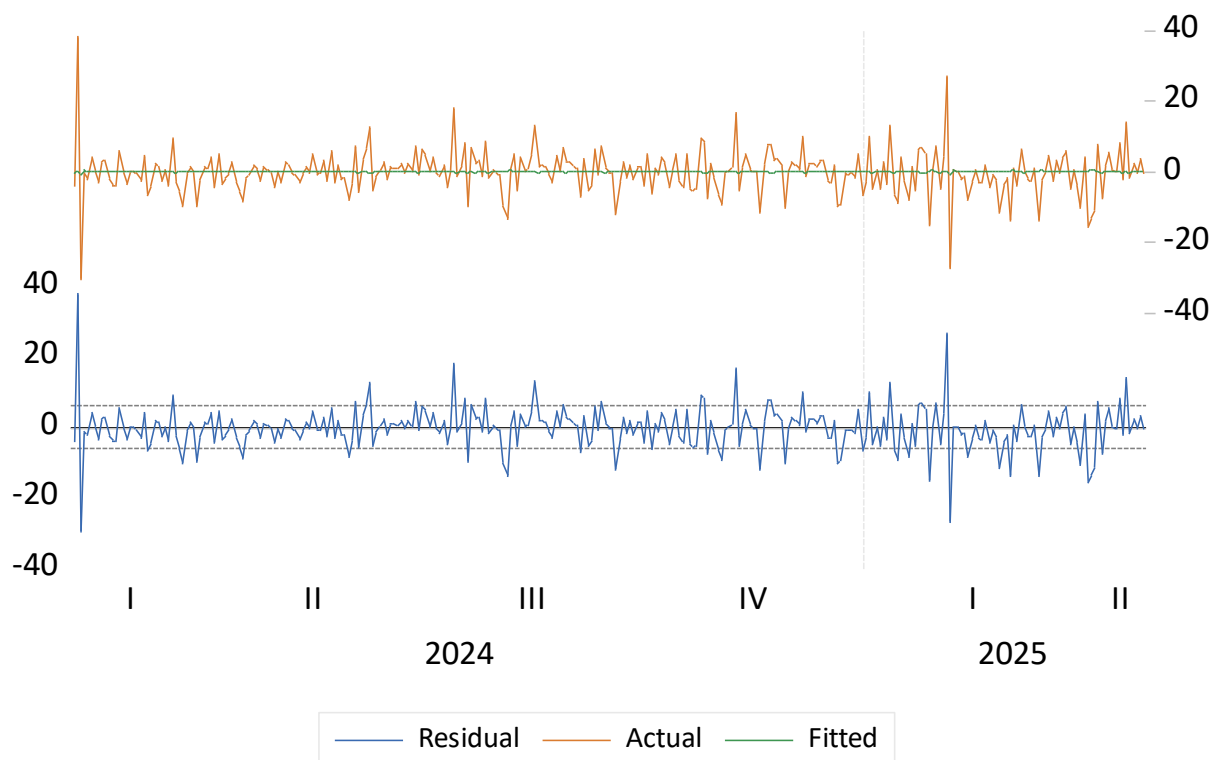
Looking at $WGT_RESID^2(-1)$, we can say past squared residuals have affected current residuals.

Figure 16: Static forecast



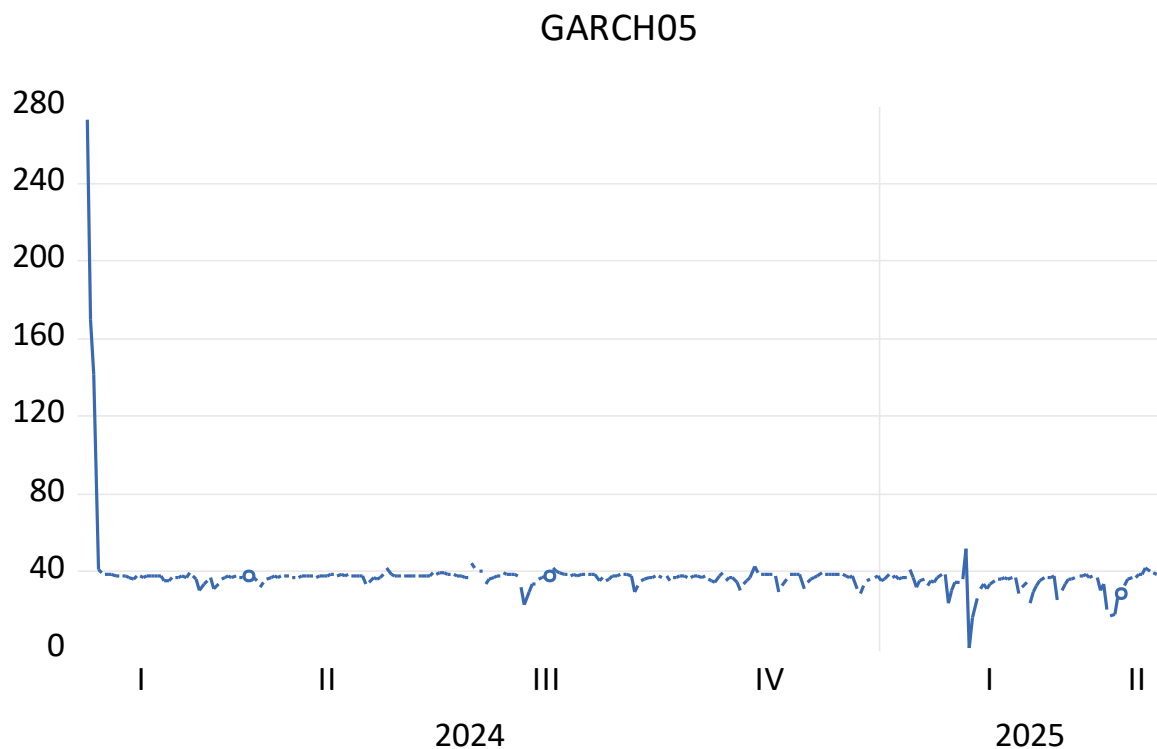
(Authors Calculation)

Figure 17: Residuals



(Authors Calculation)

Figure 18: TARCH Variance



(Authors Calculation)

Figure 18 shows the graph of TARCH variance that was applied to NIFITETF data from 1st January, 2024 to 30th April, 2025. As we can see the model adjusted from the starting point of volatility, which is shown in the graph in the starting with a steep decline. Throughout the time period, the values were generally constant, apart from the times when there were periods with higher volatility. The pattern thus we have obtained in E Views, support the use of TARCH modelling which catches volatility with asymmetry.

4.6 Estimating EGARCH model

EGARCH is an asymmetric volatility model which is improvement of the GARCH model, as it takes into account the leverage effect. It features include: Asymmetric Volatility Response, Logarithmic Formulation, there is no need for negative constraints, and it captures long memory effects in volatility.

Table 10:

Variables	Coefficients	Standard Error	z-Statistics	Probability
C	0.215421	0.294721	0.730930	0.4648
AR (1)	-0.068644	0.060699	-1.130883	0.2581

(Authors Calculation)

In the above result, C (Constant) is 0.215 with p value of 0.46.

The value for AR (1) is -0.06644 and probability value is 0.25. The coefficient is not significant, which means past returns have had impact on current return price.

4.6.1 Residual diagnostic

To determine whether or not the EGARCH model has adequately represented the volatility dynamics found in the time series data, residual diagnostics are crucial. Diagnostic tests including the ARCH LM test, sign bias test and normality test to ensure that no additional patterns are present in residuals because EGARCH models takes into account asymmetries and leverage effects in financial returns. The EGARCH model has been properly described and is adequate for precise volatility forecasting and risk analysis if the residuals resemble white noise, exhibits no auto correlation, heteroskedasticity, or asymmetry.

Figure 19: Correlogram Q Statistics

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob*
		1 -0.025	-0.025	0.1967	
		2 0.027	0.026	0.4225	0.516
		3 0.090	0.092	2.9428	0.230
		4 0.001	0.005	2.9428	0.401
		5 -0.082	-0.087	5.0177	0.285
		6 0.017	0.004	5.1098	0.403
		7 -0.033	-0.027	5.4428	0.488
		8 0.019	0.032	5.5517	0.593
		9 0.026	0.028	5.7652	0.674
		10 0.062	0.062	6.9993	0.637
		11 -0.027	-0.029	7.2395	0.703
		12 0.066	0.052	8.6496	0.654
		13 -0.031	-0.034	8.9587	0.706
		14 -0.048	-0.046	9.6930	0.719
		15 0.003	0.003	9.6955	0.784
		16 0.050	0.056	10.505	0.787
		17 0.011	0.037	10.546	0.837
		18 0.013	-0.001	10.603	0.876
		19 0.081	0.066	12.732	0.807
		20 0.058	0.052	13.838	0.793
		21 -0.028	-0.025	14.098	0.825
		22 0.121	0.105	18.913	0.591
		23 0.005	0.016	18.920	0.650
		24 -0.042	-0.035	19.518	0.671
		25 0.041	0.026	20.084	0.692
		26 0.002	0.001	20.085	0.742
		27 -0.061	-0.046	21.358	0.723
		28 0.023	0.000	21.543	0.760
		29 0.069	0.071	23.160	0.725
		30 -0.065	-0.052	24.596	0.699
		31 0.034	0.017	24.991	0.725
		32 0.097	0.075	28.203	0.611
		33 -0.001	0.028	28.204	0.659
		34 -0.010	-0.025	28.240	0.703
		35 0.047	0.022	29.022	0.710
		36 -0.060	-0.043	30.265	0.696

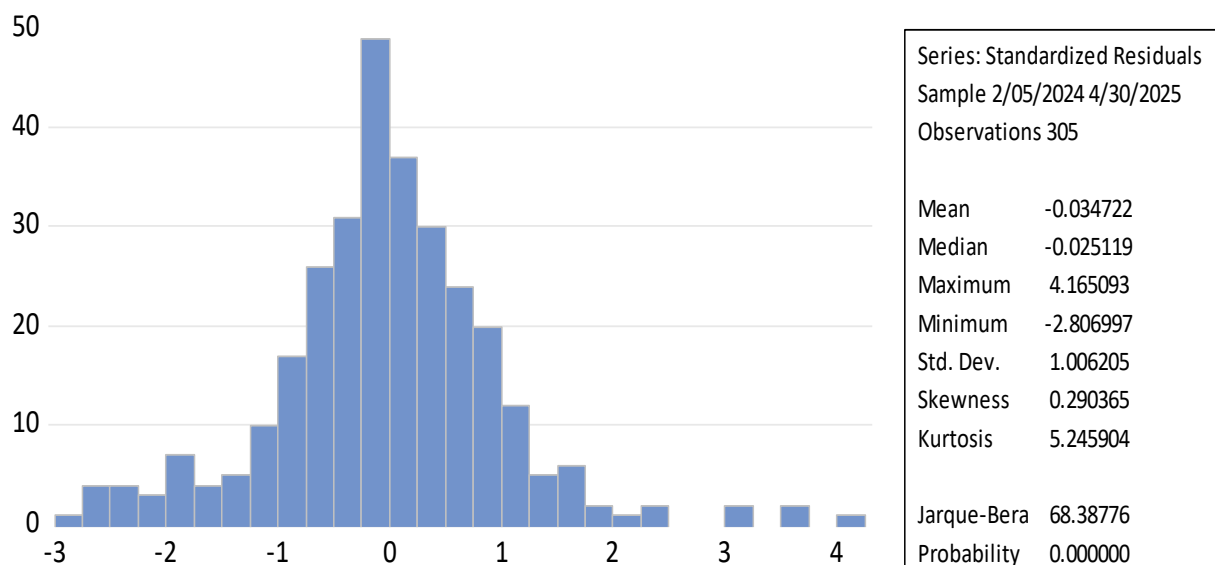
(Authors Calculation)

Figure 20: Correlogram Squared Residuals

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob*	
		1	0.096	0.096	2.8393	0.092
		2	0.014	0.005	2.9040	0.234
		3	-0.036	-0.038	3.3075	0.347
		4	-0.010	-0.003	3.3357	0.503
		5	0.022	0.024	3.4863	0.625
		6	0.012	0.006	3.5296	0.740
		7	-0.032	-0.035	3.8443	0.798
		8	0.002	0.010	3.8457	0.871
		9	0.076	0.077	5.6531	0.774
		10	-0.044	-0.062	6.2587	0.793
		11	0.092	0.102	8.9614	0.625
		12	-0.062	-0.075	10.204	0.598
		13	-0.062	-0.054	11.422	0.575
		14	0.078	0.098	13.392	0.496
		15	0.071	0.052	15.006	0.451
		16	-0.017	-0.038	15.096	0.518
		17	0.026	0.034	15.310	0.573
		18	-0.028	-0.024	15.564	0.623
		19	-0.009	-0.006	15.590	0.684
		20	-0.000	-0.022	15.590	0.742
		21	-0.079	-0.053	17.626	0.673
		22	-0.009	-0.001	17.652	0.726
		23	0.060	0.058	18.843	0.710
		24	0.005	-0.003	18.851	0.760
		25	0.055	0.040	19.881	0.753
		26	0.034	0.019	20.265	0.779
		27	-0.034	-0.011	20.657	0.802
		28	-0.044	-0.056	21.300	0.813
		29	0.016	0.026	21.383	0.845
		30	-0.002	0.005	21.385	0.875
		31	0.047	0.032	22.147	0.878
		32	-0.028	-0.031	22.407	0.896
		33	-0.012	0.001	22.459	0.917
		34	0.065	0.045	23.934	0.900
		35	-0.069	-0.070	25.601	0.877
		36	-0.032	-0.010	25.968	0.891

(Authors Calculation)

Figure 21: Histogram normality test



(Authors Calculation)

So, from the above table we can infer that:

Median value is -0.025, indicating the central value of the data series.

Maximum value is 4.165 and Minimum value is -2.806 which shows highest and lowest value in the series.

Deviation from the average value of the series is 1.00625

Value of skewness is 0.29, signifying that data may be slightly asymmetric.

Value of kurtosis is 5.245904, denoting that the distribution produces fewer and less extreme outliers as compared to the normal distribution.

Jarque-Bera is a test for ensuring the normal distribution. Returns do not follow a normal distribution, rejecting zero skewness and kurtosis.

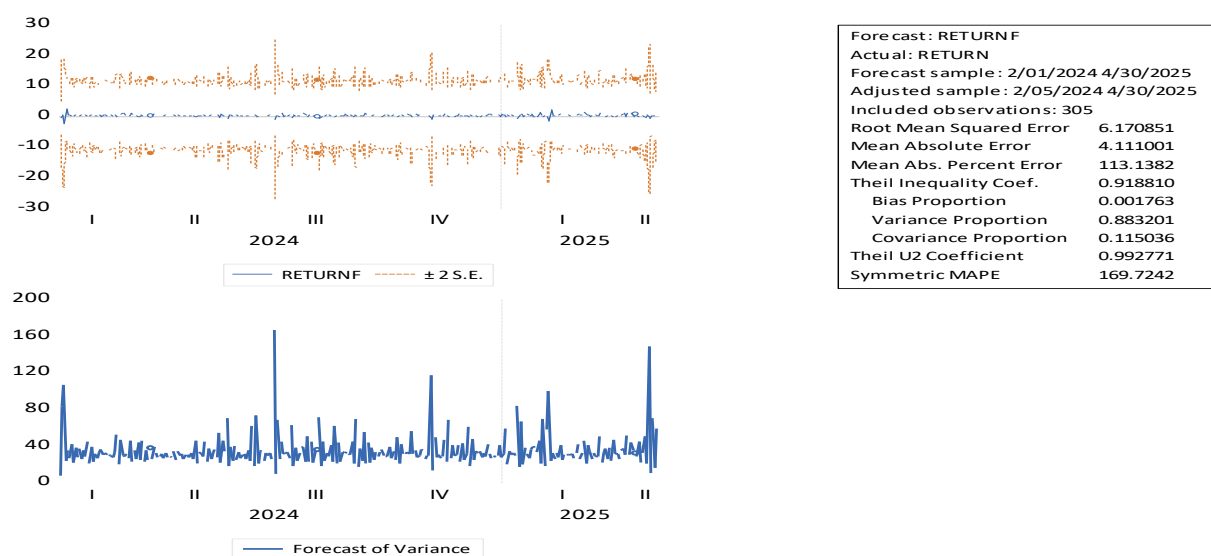
Table 11: Arch LM test

Variables	Coefficient	Standard Error	t-Statistics	Probability
C	0.911184	0.132223	6.891287	0.0000
WGT_RESID ^2 (-1)	0.096087	0.057293	1.677108	0.0946

(Authors Calculation)

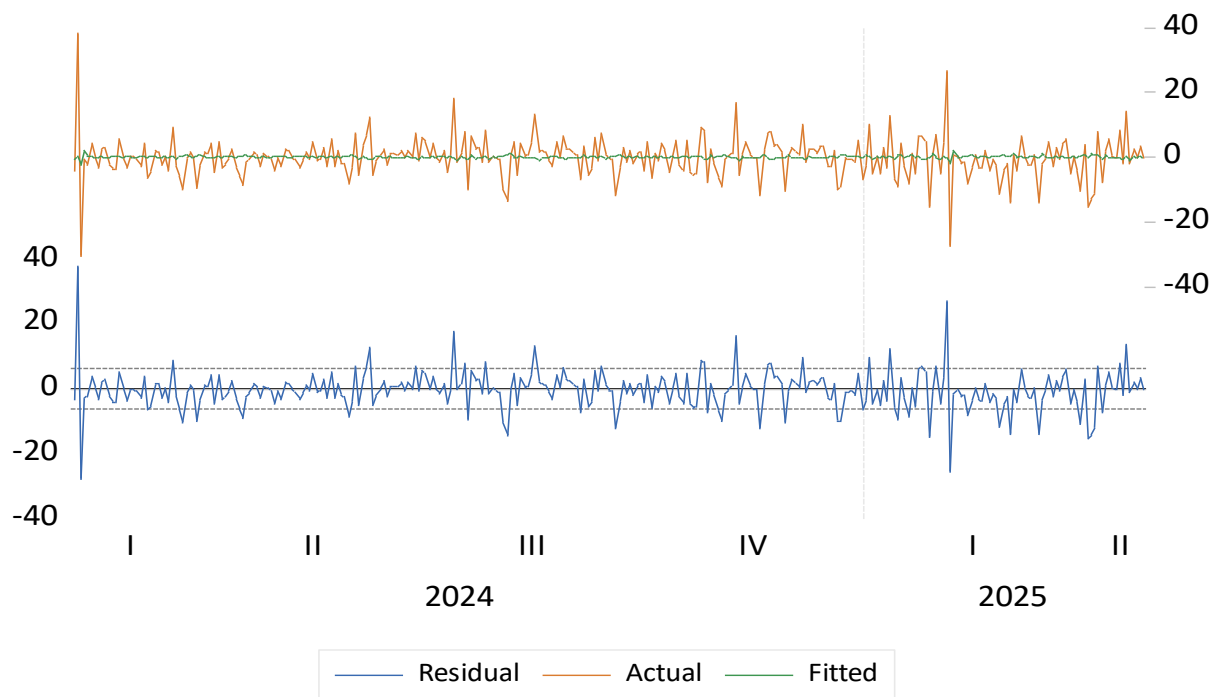
The p values being greater than 5% significance level, suggest weak evidence of ARCH effects in the residuals, which means there might be presence of volatility clustering, but not that strong, and an EGARCH model is justified but not fully.

Figure 21: Static forecasting



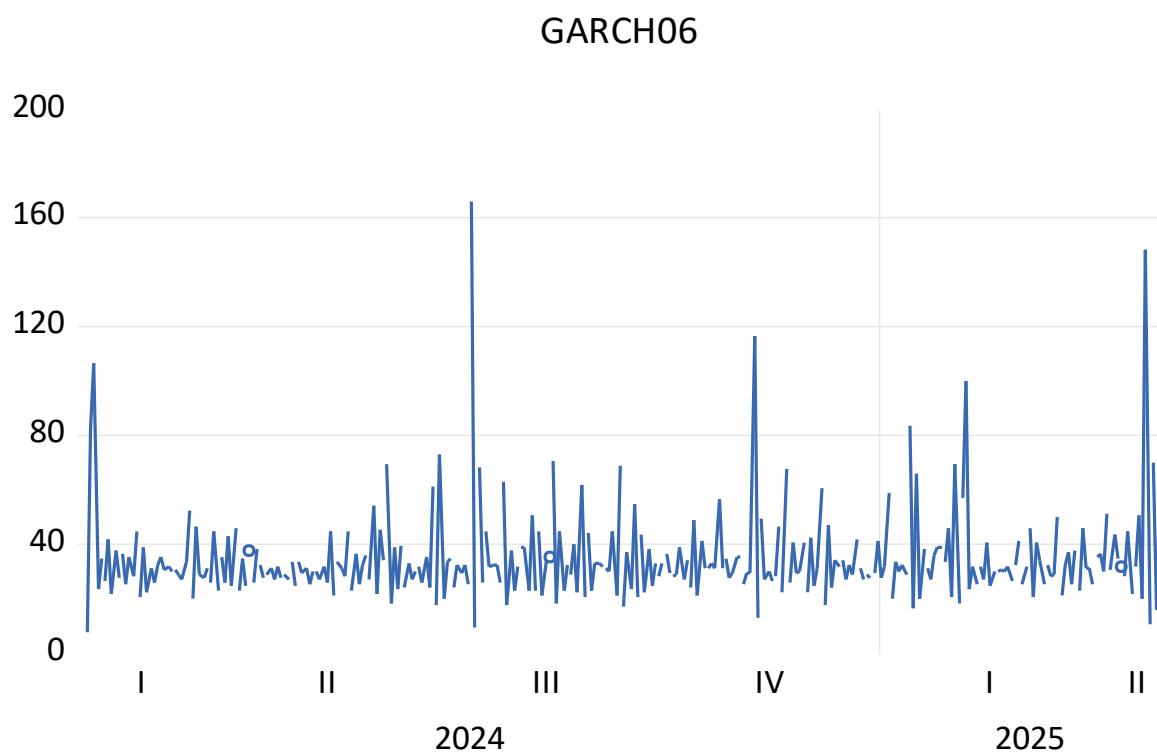
(Authors Calculation)

Figure 22: Residuals



(Authors Calculation)

Figure 23: EGARCH Variance



(Authors Calculation)

CHAPTER – 5

CONCLUSION

5.1 Conclusion

This study examines the closing price of the ETF called NIFITETF, and derive its return value. Then we tried to estimate the data using different volatility tests, namely, ARCH, GARCH, EGARCH, TARCH, etc. GARCH model was used to estimate or calculate leveraged effect.

We took daily data of the closing price of the ETF from 1st January, 2024 to 30th April, 2025 which means more than a year's data to check for the volatility of ETFs.

The data we analysed showed that daily return values were not normal, but with ARCH effect in the residual. Then we did the GARCH (1,1) model, followed by EGARCH and TARCH.

5.2 Limitation of the study

Due to time constraint of the Masters course, and other academic tasks that had to be completed, this study only took into account one particular ETF, that too for one year's data.

5.3 Scope of Further Research

This study might have extended in its estimation models and had incorporated many models like PARCH, FIGARCH, ARCH-M, NGARCH, etc.

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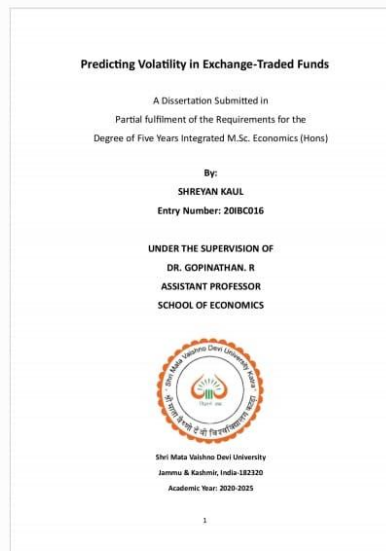


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